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Editorial Office
Quantitative Finance

Dear Editors,

We submit the manuscript “Geometric Observables for Financial Regime Detection” for consideration as an original research article in *Quantitative Finance*.

The paper introduces four geometric detectors—Berry Phase Rate, Spectral Entropy, Reduced State Purity, and Hamiltonian Sensitivity—derived from a learned spectral embedding of equity-index returns, and evaluates them against 46 classical and machine-learning baselines (including the Absorption Ratio, Hamilton’s Markov-switching model, GARCH, Bayesian online changepoint detection, and supervised Random Forests) on 17 historical crises spanning 2000–2024. Under walk-forward nested hyperparameter selection on nine labelled windows, the Berry Phase Rate achieves an unbiased out-of-sample Cohen’s $d = 0.72$ (95% percentile-bootstrap CI [0.34, 1.18]) and reduces the false-alarm rate by approximately 67% relative to a label-supervised Random Forest (1.2 vs. 3.6 alarms per year). Geometric and classical channels are largely uncorrelated ($|\rho| \approx 0.13$), suggesting the observables provide a complementary risk-overlay signal rather than a substitute for existing detectors.

Although the construction borrows formalism from quantum geometry, every computation runs on classical hardware; the contribution is empirical and methodological, not physical. Statistical evaluation uses Friedman tests with the Iman–Davenport correction, Nemenyi post-hoc analysis, and percentile bootstrap confidence intervals. All experiments are reproducible from a public code release with pinned random seeds and canonical result files.

For full transparency, a preprint of this manuscript is publicly available on SSRN (abstract ID 6685298); a corresponding arXiv posting is forthcoming. The work is original, has not appeared in any peer-reviewed venue, and is not under consideration by any other journal. We have no competing interests to declare.

Sincerely,

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